

File Permissions Management in Linux

Introduction to File Permissions

Linux file permissions determine who can read, write, or execute files and directories. Each file and directory has three levels of permission:

- **Owner (User):** The creator of the file.
- **Group:** Users belonging to the assigned group.
- **Others:** All other users on the system.

Permissions are represented as:

- **Read (r or 4)** – View file contents.
- **Write (w or 2)** – Modify file contents.
- **Execute (x or 1)** – Run scripts or programs.

To check file permissions, use:

```
ls -l filename
```

Output example:

```
-rwxr--r-- 1 user group 1234 Mar 28 10:00 myfile.sh
```

Changing Permissions with chmod

Using Symbolic Mode

Modify permissions using symbols:

- Add (+), remove (-), or set (=) permissions.

Examples:

```
chmod u+x filename # Add execute for user
```

```
chmod g-w filename # Remove write for group
```

```
chmod o=r filename # Set read-only for others
```

```
chmod u=rwx,g=rx,o= filename # Set full access for user, read/execute for group, and no access for others
```

Using Numeric (Octal) Mode

Each permission has a value:

- Read (4), Write (2), Execute (1).

Examples:

```
chmod 755 filename # User (rwx), Group (r-x), Others (r-x)
```

```
chmod 644 filename # User (rw-), Group (r--), Others (r--)
```

```
chmod 700 filename # User (rwx), No access for others
```

Changing Ownership with chown

Modify file owner and group:

```
chown newuser filename # Change owner
```

```
chown newuser:newgroup filename # Change owner and group
```

```
chown :newgroup filename # Change only group
```

Recursively change ownership:

```
chown -R newuser:newgroup directory/
```

Changing Group Ownership with chgrp

```
chgrp newgroup filename # Change group
```

```
chgrp -R newgroup directory/ # Change group recursively
```

Default Permissions: umask

umask defines default permissions for new files and directories. Check current umask:

```
umask
```

Set a new umask:

```
umask 022 # Default: 755 for directories, 644 for files
```

Conclusion

Understanding file permissions is essential for system security and proper file management. Using chmod, and chown, you can control access to files and directories efficiently.